

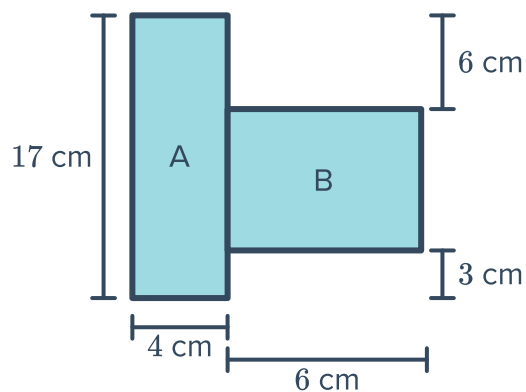
Compound shapes



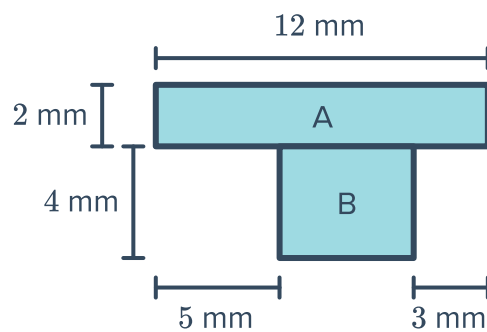
1 For each of the following figures:

- i Find the area of rectangle B.
- ii Find the total area of the compound shape.

a

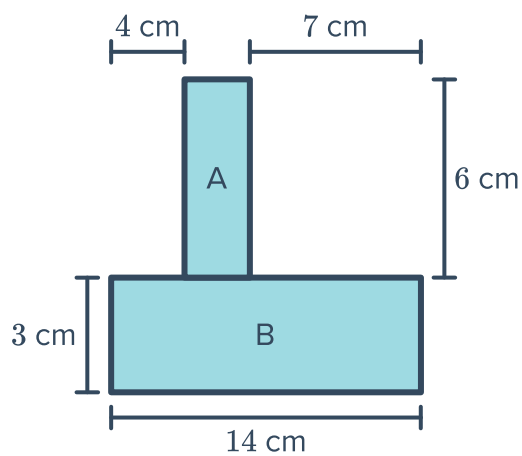


b



2 For the following compound shape:

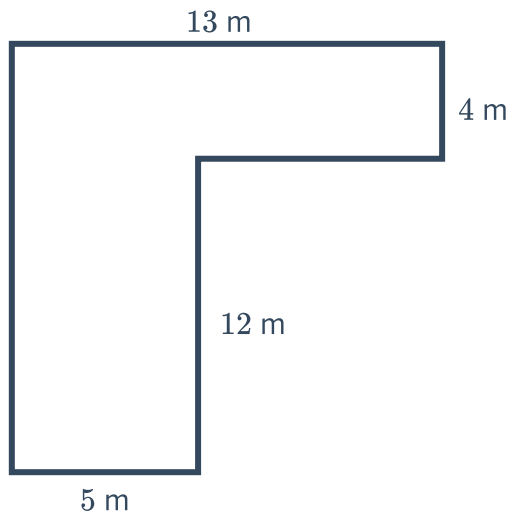
- a Find the area of rectangle A.
- b Find the total area of the given shape.



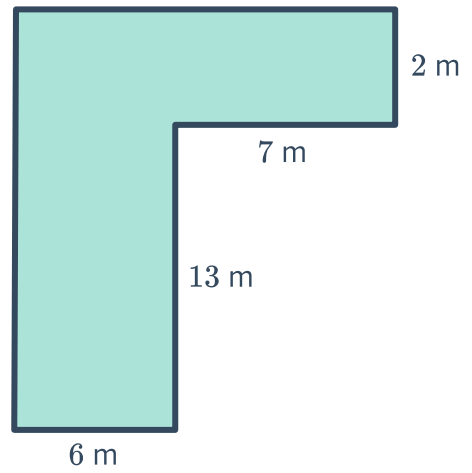
3 Find the area of the following compound shapes:



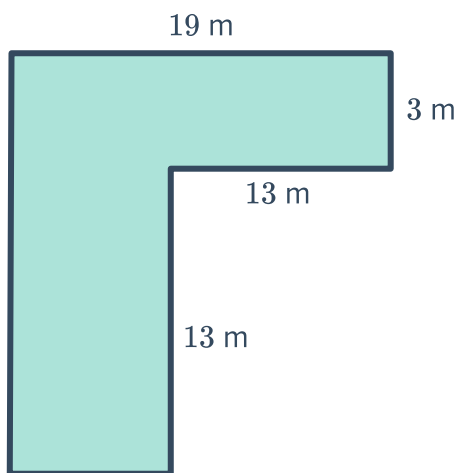
a



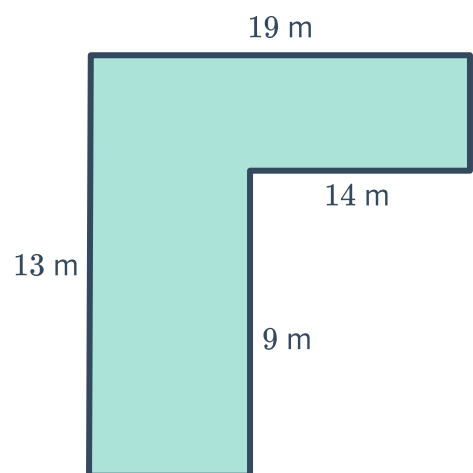
b



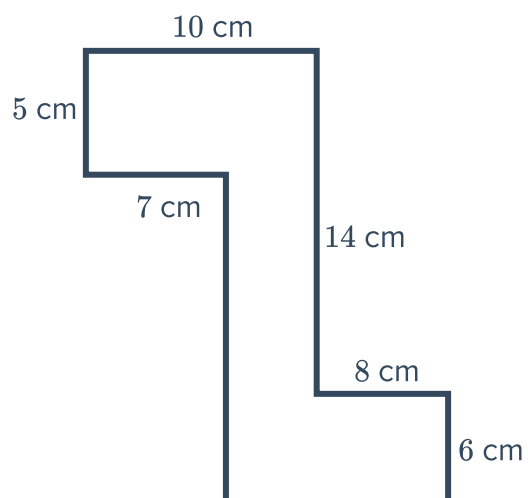
c



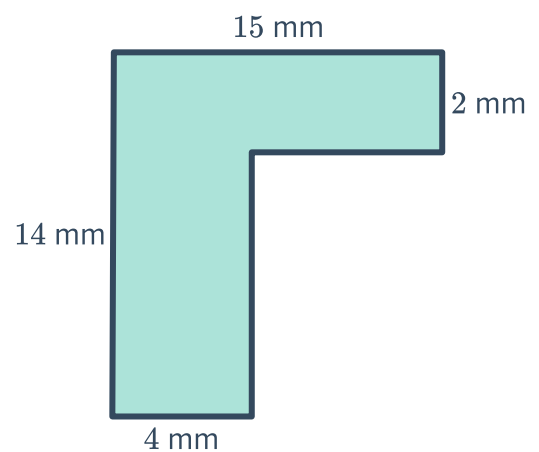
d



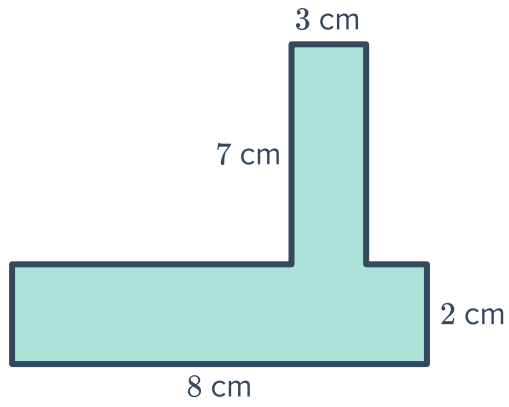
e



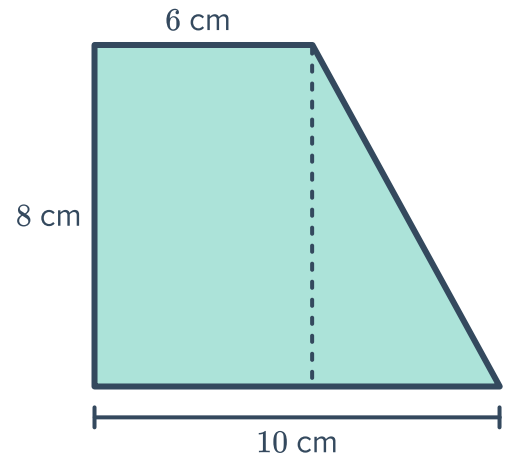
f



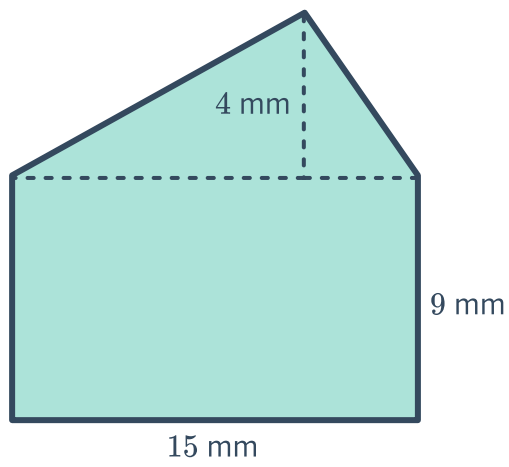
g



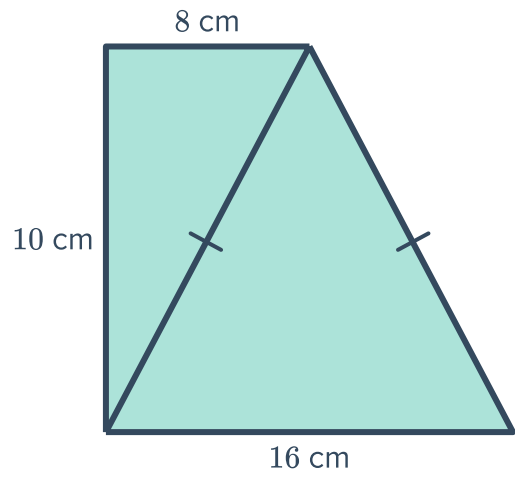
h



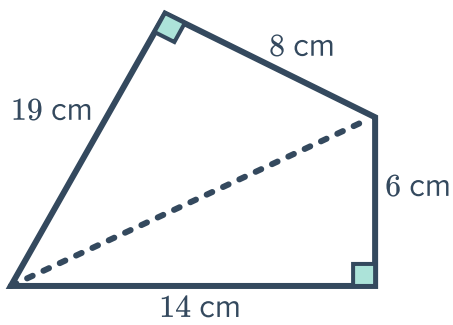
i



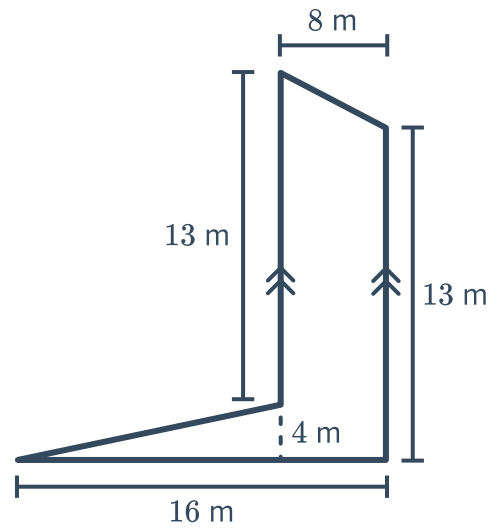
j



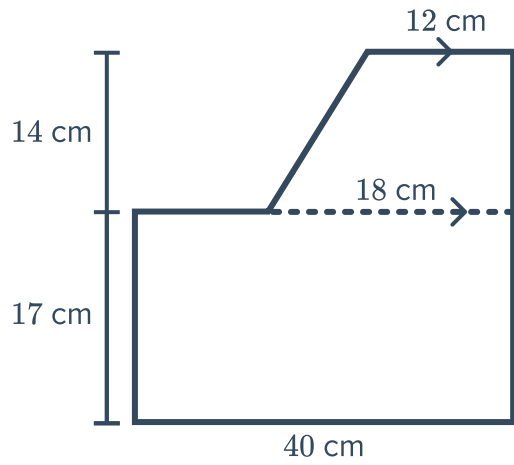
k



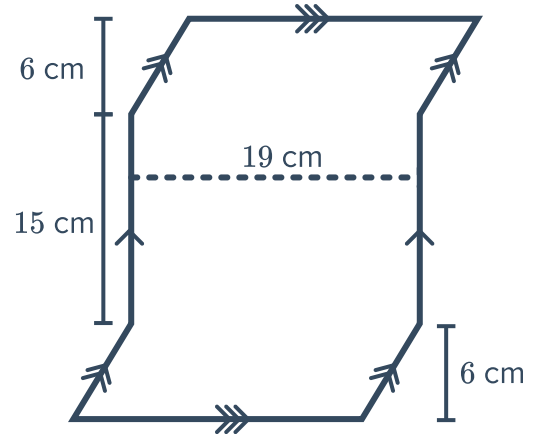
l



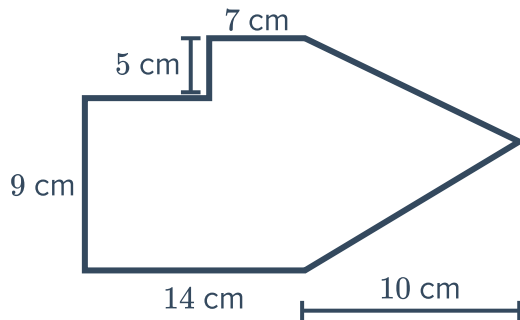
m



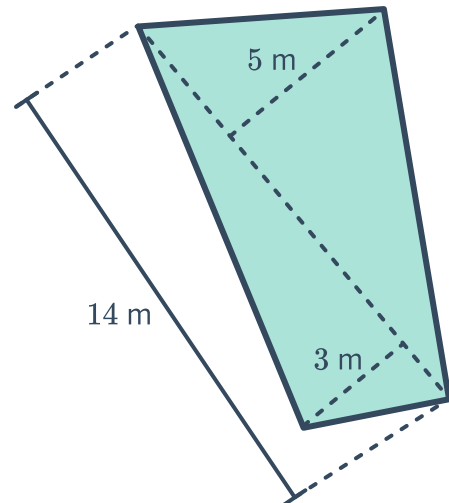
n



o



- 4 The quadrilateral in the diagram has a diagonal measuring 14 m.
Calculate the total area of the quadrilateral.

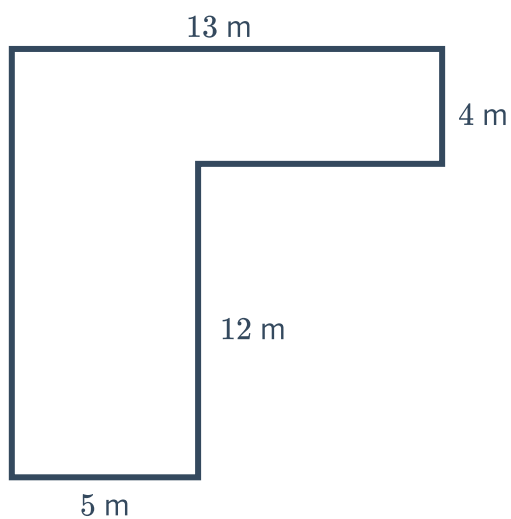


5 For each of the following compound shapes

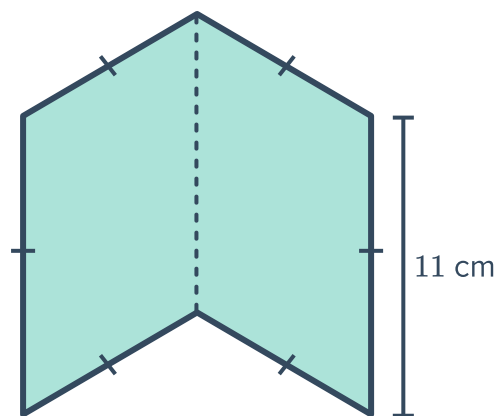


- i State the basic shapes that make up the compound shape.
- ii Find the perimeter.

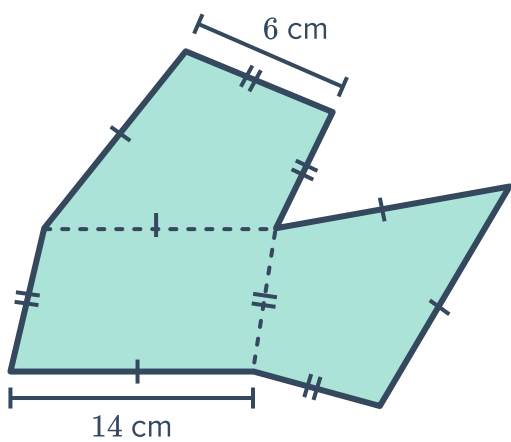
a



b



c

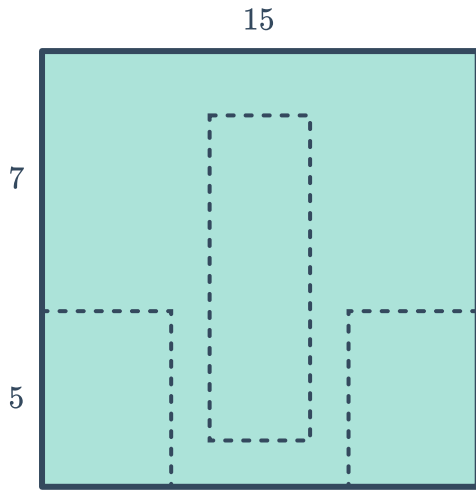




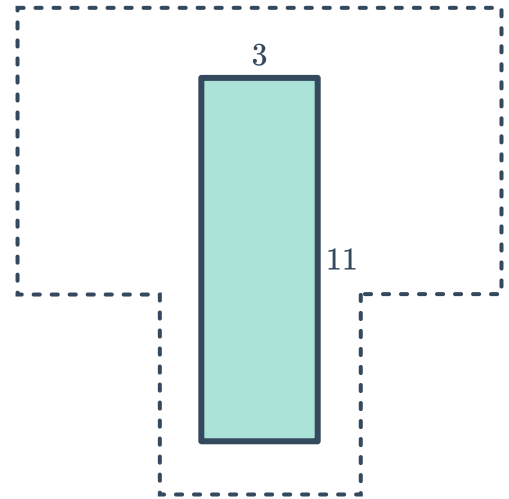
Holes and cutaways

6 Consider the following figures, where all measurements are in centimetres:

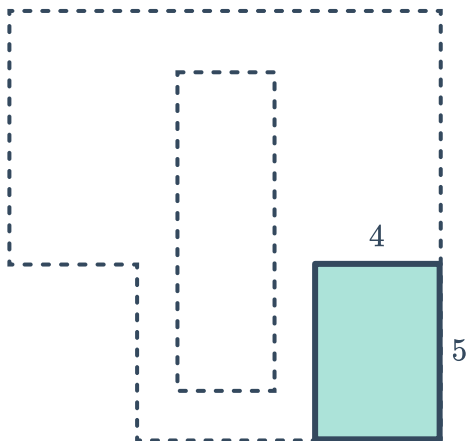
- Figure 1:



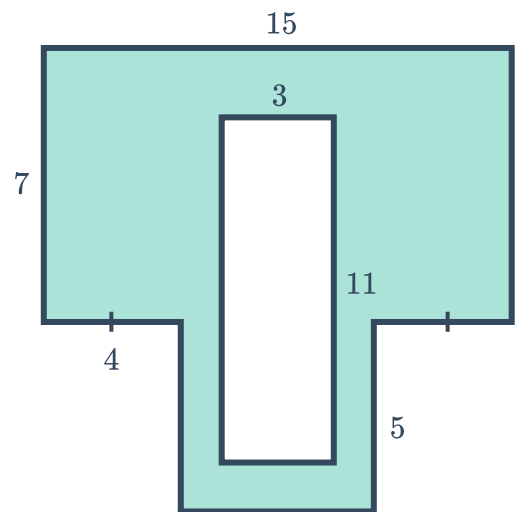
- Figure 2:



- Figure 3:



- Figure 4:

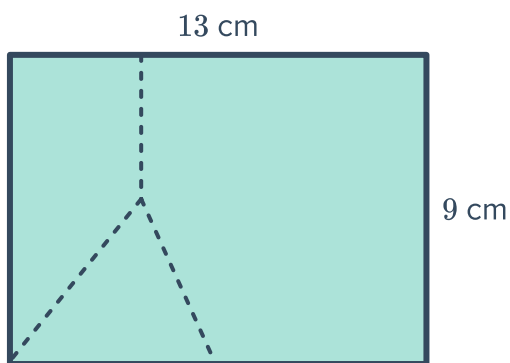


- Find the area of the entire rectangle shaded as shown in figure 1.
- Find the area of the interior rectangle shaded as shown in figure 2.
- Find the area of the corner rectangle shaded as shown in figure 3.
- Hence, calculate the area of the shaded region as shown in figure 4.

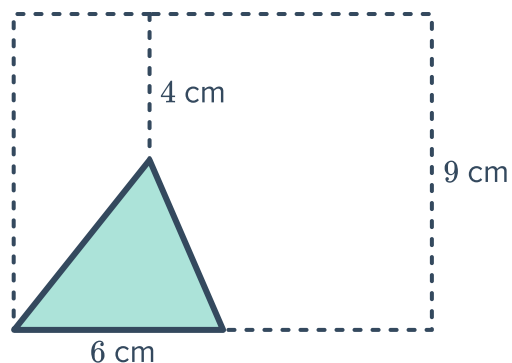
7 Consider the following figures:



• Figure 1:

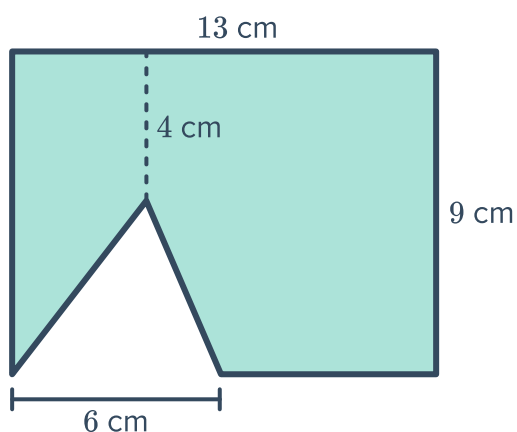


• Figure 2:



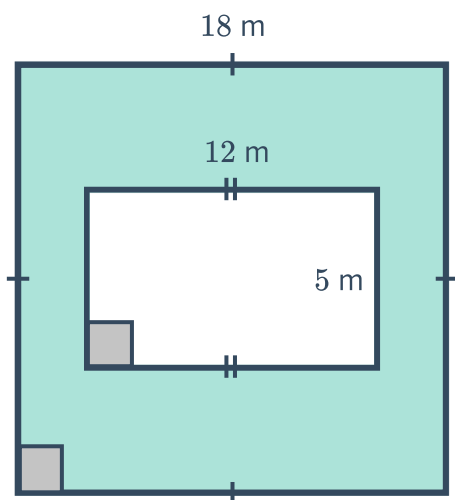
- Find the area of the shaded rectangle as shown in figure 1.
- Find the area of the shaded triangle as shown in figure 2.
- Hence, find the area of the shaded region as shown in figure 3.

• Figure 3:

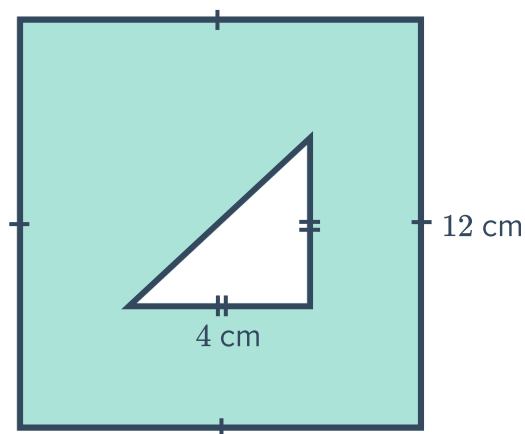


8 Find the area of the shaded region in the the following figures:

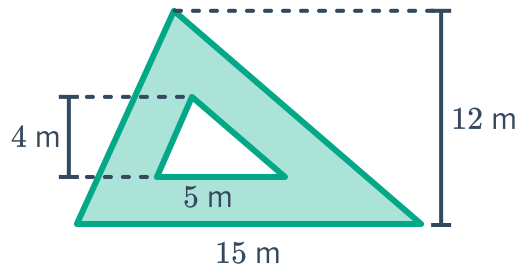
a



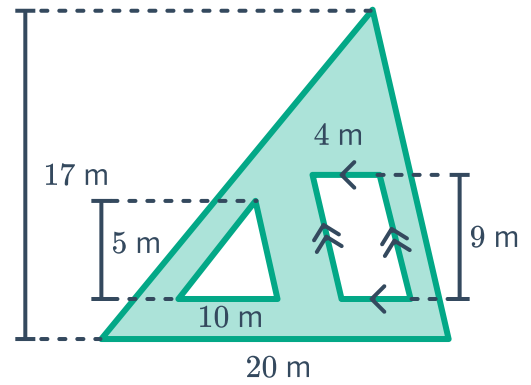
b



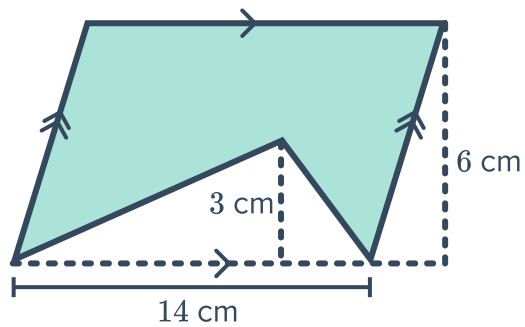
c



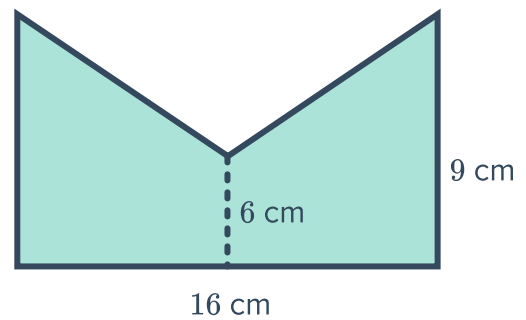
d



e

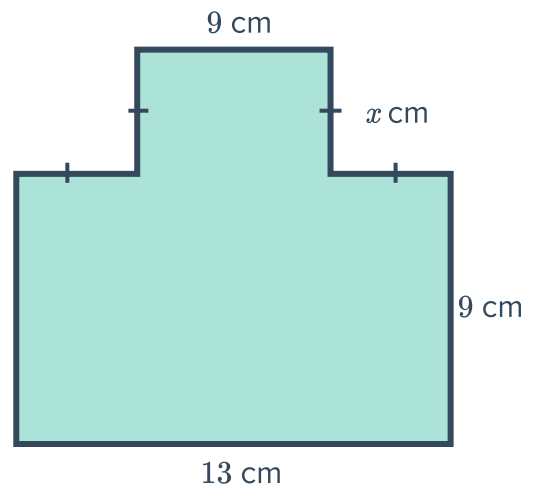


f



9 Consider the following figure:

- Find the value of x .
- Find the area of the shaded region.

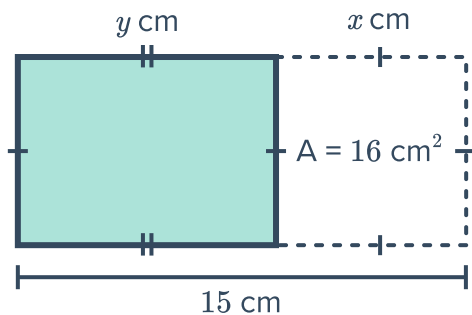


10 For each of the following figures:

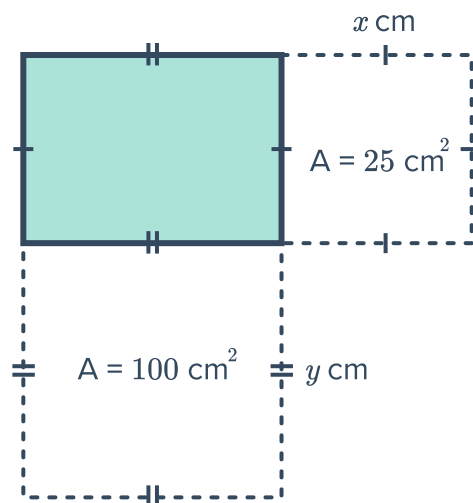


- i Find the value of x .
- ii Find the value of y .
- iii Calculate the area of the shaded region.

a

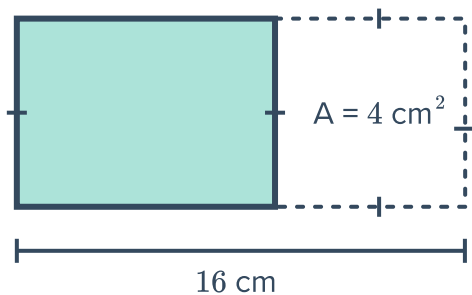


b

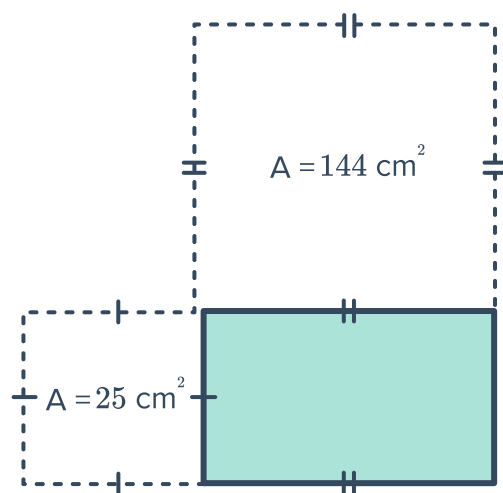


11 Find the area of the shaded region in the following figures:

a



b

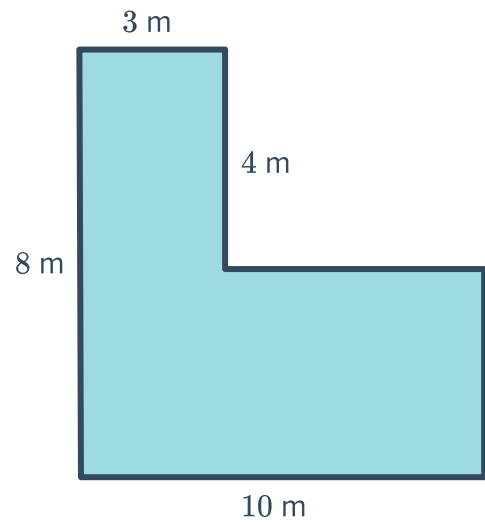




Applications

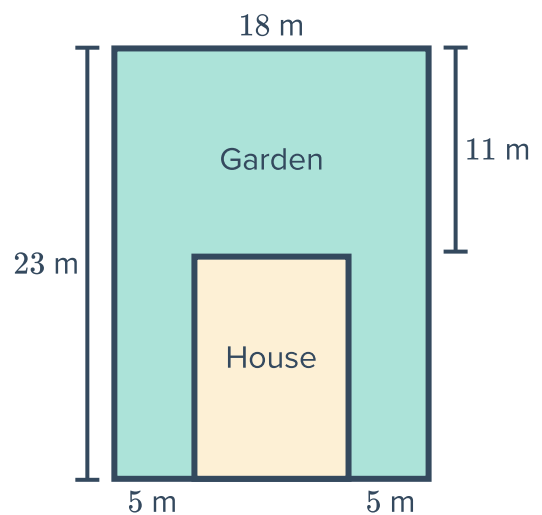
- 12** Katrina has a garden bed that is shaped as shown in the following figure:

Find the area of the garden bed.



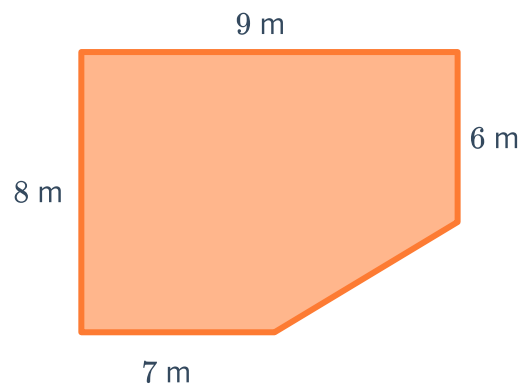
- 13** Lachlan draws the plot of land which contains his house and garden.

- a** Find the total area of the plot of land.
- b** Find the area covered by the house.

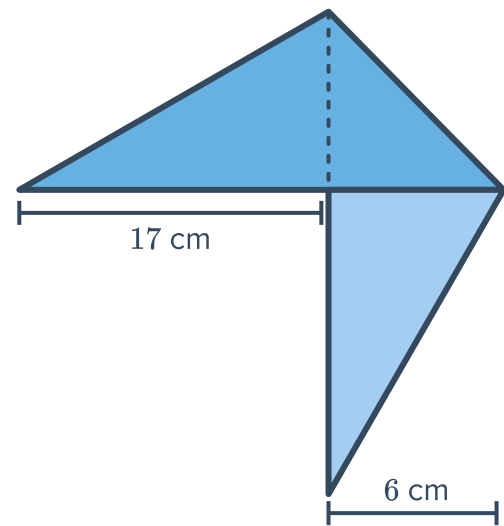


- 14** The dining room floor shown is to be tiled with slate tiles. The cost of tiling the room is \$10 per square metre.

- a** Calculate the area to be tiled.
- b** How much will it cost to tile the room?



- 15 A piece of paper in the shape of a parallelogram is folded along its shortest diagonal, as shown. Find the total area covered by the folded paper.



- 16 A farmer is going to fertilise his paddock, which has the following shape:

- Find the area of the paddock.
- If it takes 150 kg of fertiliser to fertilise 100 square metres, how many kilograms of fertiliser should the farmer purchase?

